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Research Highlights

1. We propose a VMD-CNN model to forecast the risk movement in the major crude oil markets.
2. CNN model is used to extract and model the spatio-temporal data feature in the crude oil risk changes.
3. Our model has the ability to capture multiscale risk structure using VMD model.
4. Our model produces Value at Risk forecasts at higher level of reliability.

Risk Forecasting in the Crude Oil Market: a Multiscale Convolutional Neural Network Approach

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Abstract

As the crude oil price movement is influenced by increasingly diverse range of risk factors in the crude oil markets, the crude oil price exhibits more complex nonlinear behavior and poses higher level of risk for investors than ever before. To model the crude oil risk changes at higher level of accuracy, we proposed a new multiscale approach to estimate Value at Risk. It takes advantage of Variational Mode Decomposition (VMD) model to extract and model the main risk factor in the multiscale domain, where the individual characteristics of these risk factors are modeled using ARMA-GARCH models. The Convolutional Neural Network (CNN) based nonlinear ensemble model is employed to aggregate these risk forecasts as the ensemble members and produce the optimal ensemble forecasts. Empirical evaluation of the performance of the proposed model has been conducted using the extensive dataset, constructed with daily price observations in the major crude oil markets. Experiment results confirm that the proposed risk forecasting models produce an improved forecasting accuracy for the typical risk measures such as Value at Risk.

Keywords: Crude oil risk forecasting, Variational Mode Decomposition, Deep Learning model, Convolutional Neural Network model, Value at Risk

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1. Introduction

As one of the most important commodities affecting the world economy, the crude oil receives the continuous attention from academics and industries. The movement and significant fluctuations of its trading price is being watched closely by government, industry and academics. For example, the US energy information administration releases monthly and quarterly forecasts. Even slight changes in the crude oil price could have significant impacts on the budget decisions in daily life, operation management decisions in the company, and the strategic decisions in the government (Miao et al., 2017). The crude oil price is sensitive to the unexpected market shocks and sector stocks such as the supply and demand, geopolitical factors, technological development, development of substitute energy sources, etc. (Miao et al., 2017). As a result, we have witnessed significant price fluctuation and risks in the market trading.

Facing the increasing level of risk, researches on the risk analysis and forecasting have attracted increasing research attention. Numerous researches have demonstrated that the crude oil risk is closely influenced by very complex external factors, often demonstrating multiscale data characteristics. For example, Ji et al. (2018) used CoVaR model to identify the structure break point in the dependence relationship crude oil and exchange rates. Sun et al. (2018) conducted the scale by scale analysis of the co-movement of crude oil price and the economic policy uncertainty in G7 countries. The time scale dependent characteristics has been found as a result. Chen et al. (2019) found that the oil price shocks have significant positive impacts on EPU. The relationship is time-varying over different time scales. Ji et al. (2019) showed that the investor sentiment types contribute to the WTI return variations. In Ma et al. (2019), the analysis based on the generalized dynamic factor model revealed that the underlying driving factors for the crude oil price volatility. Their empirical results suggest that the oil market is increasingly financialized when the financial factor has the biggest influence. Besides, another factor would be events that have major impacts worldwide. The impact of risk factors in the crude oil market is ultimately reflected in the empirical profit and loss distribution. There are different risk measures developed in the literature, including the standard deviation and downside risk measures such as the Value at Risk (Dowd, 2005). Compared to other risk measures, Value at Risk summarizes different risk factors into a single statistic number, that reflects concisely the downside risk level at the chosen confidence level and investment time horizon (Dowd, 2005). Researches mainly uses VaR as the risk measure to analyze the risk spillover and co-movement between the crude oil market and other financial markets. For example, Mensi et al.

(2017) have used the VaR to measure the risk level during their analysis of the asymmetric risk spillover between the crude oil market and the stock market. Du and He (2015) analyzes the extreme risk spillover between the crude oil future market and US stock market, using VaR to measure the market risk level. Ji et al. (2018) analyzes the risk spillover between the crude oil market and the exchange markets. The systemic risk is measured using CoVaR, which is calculated based on VaR. Li and Wei (2018) analyzes the dependence and risk spillover between the crude oil market and Chinese stock market, using VaR and CoVaR to measure the risk level. Aloui et al. (2014) uses the copula-GARCH approach to model the conditional dependence structure between the crude oil and gas markets. They used the Monte Carlo method to calculate copula-GARCH based VaR as the risk measure, and used the positive performance improvement to verify the effectiveness of the proposed model. Results from the aforementioned approaches are ultimately dependent on the accuracy of the risk measure.

The accuracy of VaR measure is challenged by the volatile crude oil market with transient and extreme risk factors. In the literature, VaR estimation methodology is mainly divided into the parametric and the nonparametric approaches, where both streams are being actively researched (Dowd, 2005). We have seen widespread usage of the parametric approaches such as ARMA-GARCH model, as well as the nonparametric approaches such as historical simulation and Monte Carlo simulation models (Bams et al., 2017). For example, Ghorbel and Trabelsi (2014) demonstrated that historical simulation, conditional extreme value theory and copula based models provide acceptable risk coverage when applied to energy portfolio consisting of crude oil, heating oil and natural gas. Lyu et al. (2017) found that nonlinear asymmetric GARCH model based on different distribution would produce different risk coverage. Their results favored the use of more fat tail distribution. Yu et al. (2018) combined the GARCH model, extreme value model and the vine-copula model to produce more accurate downside risk measures such as Value at Risk (VaR) and Expected Shortfall (ES) for the crude oil portfolio. Their results demonstrate the existence of long memory and leverage data features in the crude oil data. Jschke (2014) used the copula model to capture the dependence structure between WTI and Brent crude oil prices and estimate VaR and ES for the energy portfolio. Charfeddine (2016) found that FIGARCH model better captures the long range dependence in the volatility and produced the improved out-of-sample VaR forecasts. Fan et al. (2008) proposed an Extreme Value Theory based VaR calculated with Generalized Error Distribution. They found it provided superior risk forecast performance. Chiu et al. (2010) conducted a

comprehensive comparative study on different methods to estimate VaR. They found that a variant of historical simulation method performs the best while the composite method can further improve the estimation accuracy. However, forecasts from these approaches may not produce sufficient level of risk coverage when the crude oil price demonstrates the fat tail distribution and more transient behavior, facing the increasingly volatile crude oil market environment.

Recently multiscale models including the new Wavelet Analysis and Variational Mode Decomposition have been introduced to model the underlying risk structure. The downside risk forecasts in terms of VaR demonstrate the improved risk reliability and have received increasing attention (Jammazi et al., 2017; He et al., 2014; HE, 2010; He et al., 2018). When using the multiscale models to estimate VaR, the use of different parameters would result in different risk structure decomposition with different decomposition (He et al., 2009, 2015). The choice of these parameters is influenced by the market data characteristics and the noise disruptions. Forecasts produced by the multiscale models with different parameters can be viewed as the ensemble members. The nonlinear ensemble model has been proposed to aggregate the risk forecasts with different model parameters in the literature (He et al., 2009). For example, He et al. (2009) have constructed the wavelet VaR model using the neural network based nonlinear ensemble model. The deep learning model, especially the widely popular Convolutional Neural Network (CNN) model, provides a promising technique for risk estimate. It has demonstrated superior performance to traditional shallow neural network, in different natural science disciplines including the image processing, natural language processing, classification problems in the economic and finance field (Schmidhuber, 2015; Zhang et al., 2018; Litjens et al., 2017). For example, Hosaka (2019) used a series of financial ratios as the input to the CNN model to predict the corporate bankruptcy. Gunduz et al. (2017) used technical indicators as the input to CNN model to forecast the stock market movement. But since CNN model was designed originally for image processing task, it has received much less attention in time series analysis and forecasting field. Recently one dimensional CNN has been introduced in various forecasting applications (Kuo and Huang, 2018; Yan et al., 2018). Some typical examples are as follows. Wang et al. (2018) has used the CNN to forecast the individual wavelet coefficients and perform inverse wavelet transform to reconstruct the solar irradiance forecasts, with positive evidence for performance improvement. Liu et al. (2018) used the one dimensional CNN model to forecast the coefficients of the wavelet packet algorithm. Zhou et al. (2018) used one dimensional CNN model to conduct multi-step ahead forecasts.

Given the popularity of VMD models in the multiscale analysis and CNN model in the artificial intelligence research field, there have been several research attempts to combine the advantage of both models in the forecasting and classification literature. For example, work by Zang et al. (2018) has used VMD to extract modes from the photovoltaic output power data and used these converted modes as the input image data to train the multi-channel CNN models for forecasting purpose. The experiment results show the improved forecasting accuracy. Zhou et al. (2018) use VMD model to calculate a series of modes from the wind speed data and use them as input to CNN model to construct CNN based multi-step wind speed forecasting model. Experiment results show the superior forecasting performance of the proposed hybrid model compared to the performance of SVR and ELM models. Deng et al. (2019) calculated feature vectors for each decomposed modes using VMD and use CNN model to classify incipient cable failure. Experiment results show that this approach is superior to DT, KNN and BP models. San-Segundo et al. (2019) used several signal processing techniques such as Fourier analysis, wavelet analysis, and EMD to transform the EEG recording data into the input data for CNN model. The classification results show that EMD based model performs the best for focal vs. non-focal signal classification, while Fourier analysis based CNN model performs the best for seizure detection. Hasan and Bhattacharjee (2019) compressed the IMFs at the first three levels of EMD model. The compressed modes, together with other modes, are used as input to the CNN model to perform the classification of Cardiovascular disease. The experiment results confirm its superior performance. But these models usually link the multiscale decomposed data components with the future data movement or classification result directly, without the focus on certain specific data characteristics. Work in this paper is unique in that it focuses on the extraction of multiscale risk structure and data characteristics in the multiscale risk forecasts.

In this paper, we propose a new Value at Risk estimation model based on VMD model and Convolutional Neural Network models. In addition to the multiscale analysis using the VMD model, the proposed approach further takes advantage of the capability of CNN model to extract and model the hierarchical information in the risk forecasts in the multiscale domain to produce more accurate risk forecasts. The forecasts of risk information in terms of the price and volatility changes are transformed into two-dimensional image data. CNN model is used to capture this multiscale information in a data adaptive learning manner. We have applied the proposed model to the crude oil price in the major crude oil markets worldwide. The performance of the proposed model

as well as the benchmark models is evaluated in terms of the risk forecast accuracy and reliability. Significant performance improvement has been observed in terms of accuracy and reliability.

Our work contributes to the risk measurement literature with the proposition of a new model to model the multiscale risk information effectively. Our model performs the data transformation that converts one dimensional crude oil data into two-dimensional data, where multiscale statistical properties such as correlation can be identified directly. These statistical properties are not visually recognizable in the original time domain. The complex nonlinear risk dynamics is modeled using the two dimensional CNN model. CNN model is introduced to adaptively learn and model the multiscale risk structure in the VaR forecasts. VMD-CNN model has been designed specifically to capture the multiscale data characteristics in the crude oil data. Compared to the multiscale approach in the literature, VMD-CNN model takes advantage of the CNN model to further capture the multiscale and hierarchical information in the extracted modes from VMD model. In VMD-CNN model, risk forecasts at different scale are generalized into the future using the ARMA-GARCH model. These risk forecasts are used as the ensemble members for the CNN based nonlinear ensemble model to produce more stable and accurate forecasts. The combination of VMD and CNN model extracts the feature matrix with the sparsest characteristics and achieved the lowest fitting error during the training process. Due to the introduction of the deep learning models such as CNN model, it is expected to reduce the overfitting issue and improve the generalization error.

In VMD-CNN model, the multi scale model is used to explore and uncover the underlying risk structure in the crude oil market. The decomposed multiscale risk structure in VMD model as well as the multiscale risk forecasts using ARMA-GARCH model sets the multiscale framework, in which the causality relationship for the risk movement is modeled. In the meantime, we use CNN ensemble model to reduce the biases in multiscale risk forecasts, improve the robustness and accuracy of the risk forecasts in a machine learning approach. The dimensions of multiscale ensemble forecast are further reduced through a series of convolutional layers in CNN model and key ensemble forecast members are extracted and used to produce more optimal final VaR forecasts. It is a refinement to the forecasts obtained from VMD and ARMA-GARCH model. Besides, although it is known that the performance of the deep learning model is sensitive to the optimization of the hyper-parameters while the modeling process of the deep learning model remains black-box, the proposed VMD-CNN model in this paper takes advantage of the multiscale model to improve the interpretation of the deep learning model.

The rest of the paper is organized as follows. Section 2 presents the main methodology of the paper. We firstly provide a brief account of the VMD theory and CNN model. Then we explain the theory and the numerical procedure for the proposed VMD-CNN based Value at Risk estimate methodology is presented then. Results from the empirical studies using the crude oil daily data are reported and analyzed in Section 3, with in-depth analysis for further economic insights. Conclusions are drawn in Section 4.

2. Methodology

2.1. Variational Mode Decomposition

Multiscale models such as wavelet analysis and EMD model assume that there exists the multi-scale data structure in the crude oil market. They attempt to search for the correct representation of the data (Huang et al., 1998). These approaches are usually semi-parametric and adaptive in nature, which enables the extraction of data features from a mixture of diverse range of data characteristics. Among different multiscale models in the recent literature, the VMD model is a recently proposed model that takes the non-recursive approach by formulating explicitly the feature searching process as the optimization problem to identify the optimal risk factors and can deal with the well-known mode mixing problem. Variational Mode Decomposition takes the optimization approach beyond the EMD model (Dragomiretskiy and Zosso, 2014). It formulates the decomposition of the original data $x(t)$ into the modes u_k as the constrained optimization problem as in Eq. (1) (Dragomiretskiy and Zosso, 2014).

$$\min_{\{s_k\}, \{\omega_k\}} \left\{ \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) \otimes s_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \quad (1)$$

Such that $\sum_k s_k(t) = r(t)$

Where $r(t)$ is the original time series data, s_k is the decomposed IMFs, k is the number of IMFs, ω_k is the center frequency, δ is the Dirac distribution, $\otimes$ is the convolution operator.

The constrained minimization problem in Eq. (1) is transformed to the unconstrained optimization problem using the Lagrangian multiplier in Eq. (2).

$$L(\{s_k\}, \{\omega_k\}, \lambda) = \alpha \sum_k \|\partial_t[(\delta(t) + \frac{j}{\pi t}) \times s_k(t)]e^{-j\omega_k t}\|_2^2 + \|r(t) - \sum_k s_k(t)\|_2^2 + \left\langle \lambda(t), r(t) - \sum_k s_k(t) \right\rangle \quad (2)$$

Where a refers the data fidelity balancing parameter for the penalty term. It encourage the reconstruction fidelity when noises are present in the data. The unconstrained optimization problem is then solved with a series of iterative sub-optimization problem. We solve for the saddle point of the unconstrained problem in Eq. (2). Readers are referred to Dragomiretskiy and Zosso (2014) for details.

Eq.(1) and Eq.(2) defines the optimization objective and constraints for the optimization of modes during the decomposition process in VMD model, i.e. the target of the mode decomposition defined in Eq.(1) and Eq.(2) is to solve for the modes that satisfy exactly the optimization objective. Different optimization methods can be employed to find the saddle point for the unconstrained optimization in Eq. (1) and solve the constrained optimization problem in Eq. (2). The saddle point of the unconstrained problem in Eq. (2) is solved using ADMM (Alternate Direction Method of Multiplier) method (Dragomiretskiy and Zosso, 2014). The typical VMD algorithm consists of the following steps. Firstly, by applying Hilbert transform to the original data, the unilateral frequency spectrum for the original data is produced. Secondly, an exponential tuned to the center frequency of the data is added to the unilateral frequency spectrum of the mode to shift it to the baseband. Thirdly, the mode for the bandwidth is estimated using H^1 Gaussian smoothness of the demodulated data.

The optimization defined in Eq.(1) and Eq.(2) in VMD theory can be used to identify the underlying multiscale data components for the crude oil price. It can not be applied to the decomposition of the crude oil risk directly since risk is a latent variable calculated using the price. However, the optimization based decomposition process would produce a multiscale data structure from the crude oil price. Risk measure can be further defined in this multiscale data structure to construct the multiscale risk structure for the crude oil market. In the traditional EMD model, even if we forced the restriction of the maximum number of modes to decompose, the errors would still accumulated during the iterative sifting process. But in VMD model, the enforcement of the maximum number of modes to decompose serve as the additional constraints so that the errors

during the optimization process will be balanced. Since we may set the maximum number of modes based on the theoretical support in the literature, this constraint serves as the bridge to bring in the theoretical foundations for the number of modes to decompose.

2.2. Convolutional Neural Network Model

CNN model is a specialized type of feedforward neural network model with deep layers and special network design to estimate their parameters. Different from the traditional shallow neural network, CNN is designed to focus on the extraction of spatial invariant data features by utilizing the local receptive field and shared weights technique (Lecun et al., 1998). This makes it more efficient in neural network training, with much fewer parameters. Thus in CNN model, the network can get very deep while the number of parameters to train stays within the manageable range. It consists of several types of layers including convolutional layers and pooling layers, in addition to the fully connected layers (Lecun et al., 1998).

The convolution layer contains neurons that linearly convolves the input feature maps to the kernel functions, also known as the filters. The input to the first convolution layer is the original image data. The neurons in the same convolution layer share the same kernels. Through the convolution operations, each filter can generate one feature map. This feature map corresponds to the data characteristics extracted using some specific filter. Filters usually are stacked in CNN model to generate multiple feature maps so that different data characteristics are extracted and reflected in the generated feature maps using different filters (Lecun et al., 1998).

With input image data X , the feature map at layer l is calculated as in Eq. (3) (Zhao et al., 2018).

$$X^{l+1} = f(W^{l+1} \otimes dX^l + b^{l+1}) \quad (3)$$

Where W is the weight matrix, b is the bias matrix, f is the activation function for nonlinear transformation of the output data.

The activation function nonlinearly transform the input data X as in Eq. (4).

$$f(X) = \max(0, X) \quad (4)$$

Typical activation function includes sigmoid, Rectification Linear Unit (ReLU). If ReLU is used, it would take the maximum value of 0 and the X (Zhao et al., 2018).

The pooling layer contains neurons that multiply the weight matrix, known as the stride, with the feature maps. It reduces the dimension and resolution of the feature maps so that the transformed feature maps can focus on more abstract and relevant features, where the concrete details of the feature is less relevant. Typical pooling function includes the max pooling and mean pooling function, as in Eq. (5).

$$d(X) = T_{r \in R}(X_{i \times T+r}) \quad (5)$$

Where T is the maximization or mean operations.

2.3. VMD-CNN based Value at Risk Estimate

As far as the crude oil price movement is concerned, empirical studies have shown that the crude oil price demonstrates a wide range of data features such as autocorrelation, multiple regimes, multiscale data feature, etc. Recent research has demonstrated the significance of the multiscale data features in crude oil price movement (He et al., 2012a). Based on these evidences, we assume that the crude oil price contains the multi scale data characteristics. Both multi scale model and AI model can be used to capture and model the multi scale data feature.

Two multi scale models including wavelet analysis and EMD models have been used in the literature to help separate and extract the main influential factors with distinct multiscale behavioral characteristics (Wang and Wang, 2017; Qiu et al., 2017). They explicitly model the original crude oil price as the sum of a series of data components, by taking advantage of the statistical properties such as the orthogonality and sparsity. For example, EMD model has been used to model the financial data and derive the underlying IMFs (Wang and Wang, 2017; Qiu et al., 2017). In the meantime, based on different model assumptions, estimates from each multiscale model provide different level of approximation to the true multiscale data structure in the crude oil market. When the econometric models are used to project the estimated multiscale data structure into the future, the forecasts introduce the additional biases. The usual linear summation assumption to reconstruct the final forecasts may produce poor forecasts. To solve these problems, CNN model can be used. It could take advantage of the existence of multi scale data feature with a different

Artificial Intelligence approach. During the model training process, it utilizes a series of filters to extract the main data features at different layers, corresponding to different scales.

It could use a number of filters to reduce the dimensions of the main feature, and then determine the optimal set known as the feature maps. The feature map is linked to the final forecasts by the neural network set up. The parameters for the feature map and the neural network are optimized by gradient descent method to minimize the empirical risk during the training process.

Therefore, we propose a VMD-CNN model to combine both multi scale models and CNN model to better exploit the multi scale data features in the crude oil risk estimates. VMD model is used to extract the coarse level data components at different scales, and then the individual risk is estimated for these data components. CNN model is used to model the dynamic ensemble between the individual risk forecasts for the coarse level data components and the crude oil price movement in the ensemble modeling framework. The individual risk forecasts at different scales are analyzed to find the main feature during the model training process, so that the estimated dynamic ensemble relationship between these features and crude oil price risk would be more accurate.

The proposed VMD-CNN model in this paper consists of a two-stage feature extraction and modeling process.

(i) In the first stage, we assume that there exists some specific multi scale risk structure with the preset number of risk factors. To construct the multiscale risk estimates, we use the VMD model to extract the coarse level data features in the multiscale domain, where data components with distinct frequency bands across scales are extracted. The crude oil data r_t is decomposed into N number of modes as in Eq. (6).

$$r_t = \sum_{n=1}^N mode_{n,t} \quad (6)$$

Where $mode_{n,t}$ is the mode matrix. N is the pre-set maximal level of decomposition scale. For example, in this paper we assume that there are 8 risk factors for the crude oil risk fluctuations, based on the relevant literature.

We then use the ARMA-GARCH model to estimate the conditional mean μ and conditional standard deviation σ for each decomposed mode data, as in Eq. (7).

$$\begin{aligned}\widehat{\mu}_{(t,s)} &= a_{(0,s)} + \sum_{i=1}^r a_{(i,s)} r_{(t-i,s)} + \sum_{j=1}^m b_{(j,s)} \epsilon_{(t-j,s)} \\ \widehat{\sigma}_{(t,s)}^2 &= \omega_{(0,s)} + \sum_{i=1}^p \omega_{(i,s)} \sigma_{(t-i,s)}^2 + \sum_{j=1}^q \beta_{(j,s)} \epsilon_{(t-j,s)}^2\end{aligned}\quad (7)$$

where $\widehat{\mu}_{t,s}$ and $\widehat{\sigma}_{t,s}^2$ are the conditional mean and conditional variance at time t respectively. $r_{t-i,s}$ ($i = 1 \dots r$) is the lag r returns with parameter $a_{i,s}$, and $\epsilon_{t-j,s}$ ($j = 1 \dots m$) is the lag m residual in the previous period with parameter $b_{j,s}$. $\sigma_{t-i,s}^2$ ($i = 1 \dots p$) is the lag p variance with parameter $\omega_{i,s}$, and $\epsilon_{t-j,s}^2$ ($j = 1 \dots q$) is the lag q squared errors with parameter $\beta_{j,s}$ in the previous period.

The time lags are determined using the model tuning data set by minimizing the AIC information criteria. Then we convert the conditional mean forecast matrix and the conditional standard deviation forecast matrix to the gray-scale image data.

(ii) In the second stage, we assume that there is a nonlinear relationship between the conditional mean forecast matrix with the spot return, as well as conditional standard deviation forecast matrix with the empirical standard deviation. We introduce the CNN based ensemble model to forecast the risk movement. We use the CNN based nonlinear ensemble model to ensemble individual forecasts of different scales. The CNN model structure is determined using the model tuning data set.

$$\widehat{\mu}_t = f_{\mu}(\widehat{\mu}_{t,s}), \widehat{\sigma}_t = f_{\sigma}(\widehat{\sigma}_{t,s}) \quad (8)$$

Where f_{μ} and f_{σ} are the nonlinear map estimated using CNN model for the conditional mean and conditional standard deviation respectively. In the VMD-CNN model, we generalized the risk forecasts into the future at different scales. Our approach in this paper essentially constitutes an ensemble model integrating different risk forecast at different scales. The μ and σ in eq.(8) are used as the ensemble members for the ensemble forecasts for conditional mean and conditional standard deviation respectively. Each forecast at each scale is represent forecasts as the ensemble member. These ensemble members have estimation biases. Ensemble members are combined in a nonlinear approach using the Artificial Intelligence models. Neural network model has been widely used in the literature to construct the nonlinear ensemble forecasting model. For example, Yu et al. (2008) proposed a neural network based nonlinear ensemble model for foreign exchange rate movement. They found the improved forecasting performance. But the risk forecasting problem in this paper is

formulated as a CNN based nonlinear ensemble forecasting problem. CNN model is introduced to construct the nonlinear ensemble, instead of the use of neural network model. Compared to the neural network model, CNN model employs the additional feature extraction through the convolutional layers, and it reduces the overfitting issue and improves the forecasting accuracy.

With the estimated μ_t and σ_t , we calculate the ensemble forecasts. Suppose there is a portfolio with investment value X , consisting of n assets, each with returns $r_t, (t = 1, 2, \dots, n)$ and portfolio weight $\omega_t, (t = 1, 2, \dots, n)$. At the confidence level cl and the holding period h , VaR value is defined as Eq. (9).

$$VaR(h, \alpha, \omega, s) = -h\widehat{\mu_{t,\omega,s}} + \sqrt{h\widehat{\sigma_{t,\omega,s}}}Z_\alpha \quad (9)$$

3. Empirical Studies

In this paper, we use the spot price in the major crude oil markets such as West Texas Intermediate (WTI) market, Brent market and Organization of the Petroleum Exporting Countries (OPEC) market as the experiment subjects to conduct empirical studies in this paper. The original data source is Energy Information Administration, US. The data were obtained at daily frequency from 2 January 2003 to 5 August 2019 from quandl, which is a public data warehouse that redistribute the data from the original data source for convenience and ease of access for research and development (Quandl, 2018). It is known that the crude oil markets have shifted through several regimes over the years, from the relatively stable era to significantly volatile market environment since the new millennium. The market risk has been significantly higher since 2003 due to the increasingly constrained supply and global economic shifts. We choose the starting date of the sample as the beginning of 2003 because since 2002 crude oil markets shifted into the latest regime when volatile price movement and increasing market risk exposure have become the norm (He et al., 2012b). This results in 4182 observations for WTI market, 4222 observations for Brent market and 4280 observations for OPEC market. The data are used in three phases of the empirical studies, which consist of model training for estimating parameters for the ARMA-GARCH model, CNN model calibration for choosing the network structure and hyper parameters, and the model evaluation for testing the model performance out-of-sample. Thus, the data are divided into three parts accordingly to facilitate the requirement of the empirical studies. The division of the data follows

70-30 ratio, i.e. the first 49% data is used for model training, the next 21% data is used for model validation and the rest 30% is used for model evaluation (He et al., 2012b). Our experiments are implemented on the Matlab 2017b computing platform, with its new neural network toolbox with the updated ability to implement the CNN model. Experiments are run using the cloud computing service provided by Ali corp. More specifically we use the hardware configuration with Intel Xeon E5-2682 CPUs. The model performance is evaluated using standard performance measures in the risk measurement literature. (i) The number of exceedances, where the exceedance is defined as the instance that the predicted VaR value exceed the trading loss in the market. (ii) The likelihood ratio test of unconditional risk coverage (Dowd, 2005). The number of exceedances counts the number of times that the actual loss is greater than VaR. The number of exceedances for the ideal VaR model is expected to strike the balance between the model reliability and the capital efficiency. The likelihood ratio (LR) test of unconditional coverage (UC) proposed by Kupiec (1995) is a widely-used backtesting technique. It is a frequency based hypothesis approach that specifies the null hypothesis (i.e. the model is accepted at a given confidence level) and tests its statistical significance with the available sample data (Kupiec, 1995; He et al., 2016, 2018). In the unconditional coverage test, the convergence of exceedances to the theoretical value is tested. The null hypothesis is: the observed frequency of exceedance is equal to the actual frequency of exceedance., i.e. the convergence of exceedances to theoretical value (Kupiec, 1995). The likelihood ratio test statistics is constructed as in (10).

$$LR_{UC} = -2\ln[(1-p)^{n-x}p^x] + 2\ln[(1-\frac{x}{n})^{n-x}(\frac{x}{n})^x] \quad (10)$$

The test statistics would converge to the $\chi^2(1)$ distribution.

We calculated the descriptive statistics using the model training data to perform the quantitative evaluation of the data characteristics. Results are reported in table 1.

Table 1: Descriptive statistics and statistical tests

Statistics	$Mean_{\times 10^{-4}}$	Standard Deviation	Skewness	Kurtosis	p_{JB}	p_{BDS}
r_{WTI}	3.7361	0.0236	-0.0880	8.4037	0.001	0
r_{Brent}	4.0533	0.0211	0.0121	8.7175	0.001	0
r_{OPEC}	4.0694	0.0156	-0.3703	7.2649	0.001	0

We find from statistics results in table 1 that the distribution of the crude oil price in all three markets deviate from the normal distribution. The means of the return in both markets are positive. The kurtosis values in both markets are significantly larger than the standard value 3. The null hypotheses of both JB and BDS tests in all markets are rejected statistically at the 5% level of significance. In the empirical analysis literature, the crude oil prices in both markets have been found to demonstrate non-normal behavioral characteristics, over different time windows. This finding is consistent with the results reported in prior studies (He et al., 2018). Although multi-scale models are widely used in the literature, the statistical tests are not available in the literature due to the complexity of multiscale modeling. For example, there are different definitions for the scale, such as time scale, volatility scale, etc. In the meantime, the existence of the multiscale and hierarchical data features are shown in numerous empirical studies in the literature. For example, the regime switching model and the mixture distribution model have been used extensively to model different regimes and distribution in the data, which provides the indirect evidence on the existence of different components with different scale characteristics in the data. Zhang and Zhang (2015) found the empirical evidence on the existence of three regimes in the crude oil markets using the Markov regime switching model. Meade (2010) showed the statistical evidence that the crude oil price distribution can be described as Gaussian or t mixture distribution. The heterogeneous regressive models have also been extensively used. Haugom et al. (2016) uses Heterogeneous Autoregressive - Quantile Regressoin model to capture different impact of volatility at different scales. Gong and Lin (2017) found that the HAR model performed the best over the long term time scale in the crude oil market. Campos et al. (2017) found that HAR model provided the best fit and produce the best forecasts for VIX. Ma et al. (2017) showed that the introduction of regimes into the HAR model would improve the forecasting accuracy for the realized volatility for crude oil futures.

Using the validation set, we conduct the experiments to identify the optimal network structure for CNN model. We calculate MSEs for VaR estimates using different model parameters, with filter size 1 and 2, as well as the number of filters from 1 to 8. The first number in the bracket refers to the filter size while the second number refers to the number of filters. In Table 2 we present the model performance using different model parameters.

Table 2: Model performance using in-sample training data

Models	$e_{w,95\%}$	$e_{w,97.5\%}$	$e_{w,99\%}$	$\overline{e_w}$	$e_{b,95\%}$	$e_{b,97.5\%}$	$e_{b,99\%}$	$\overline{e_b}$	$e_{o,95\%}$	$e_{o,97.5\%}$	$e_{o,99\%}$	$\overline{e_o}$
(1,1)	2.3993	2.8548	3.4821	2.9121	1.8798	2.2575	2.7753	2.3042	1.1283	1.3227	1.5886	1.3465
(1,2)	2.3989	2.8543	3.4814	2.9115	1.8800	2.2578	2.7758	2.3045	1.1281	1.3224	1.5883	1.3463
(1,3)	2.3988	2.8542	3.4813	2.9114	1.8798	2.2575	2.7754	2.3042	1.1285	1.3229	1.5889	1.3467
(1,4)	2.3988	2.8542	3.4813	2.9115	1.8799	2.2576	2.7755	2.3043	1.1284	1.3228	1.5887	1.3466
(1,5)	2.3989	2.8544	3.4815	2.9116	1.8803	2.2581	2.7762	2.3049	1.1283	1.3226	1.5885	1.3464
(1,6)	2.3983	2.8535	3.4803	2.9107	1.8802	2.2580	2.7761	2.3048	1.1283	1.3226	1.5885	1.3464
(1,7)	2.3986	2.8540	3.4810	2.9112	1.8803	2.2581	2.7762	2.3049	1.1285	1.3229	1.5889	1.3468
(1,8)	2.3988	2.8541	3.4812	2.9114	1.8804	2.2582	2.7763	2.3049	1.1285	1.3229	1.5889	1.3467
(2,1)	2.3991	2.8545	3.4817	2.9117	1.8802	2.2580	2.7760	2.3047	1.1285	1.3228	1.5888	1.3467
(2,2)	2.3987	2.8540	3.4810	2.9112	1.8797	2.2574	2.7752	2.3041	1.1284	1.3227	1.5887	1.3466
(2,3)	2.3987	2.8540	3.4810	2.9113	1.8801	2.2579	2.7758	2.3046	1.1287	1.3232	1.5893	1.3471
(2,4)	2.3992	2.8548	3.4820	2.9120	1.8808	2.2587	2.7769	2.3055	1.1285	1.3229	1.5889	1.3468
(2,5)	2.4167	2.8693	3.4932	2.9264	1.8802	2.2580	2.7759	2.3047	1.1306	1.3258	1.5929	1.3498
(2,6)	2.3978	2.8529	3.4797	2.9101	1.8911	2.2708	2.7910	2.3176	1.1288	1.3232	1.5892	1.3471
(2,7)	2.5289	2.9911	3.6262	3.0487	1.8665	2.2370	2.7464	2.2833	1.1277	1.3219	1.5875	1.3457
(2,8)	2.5496	3.0166	3.6572	3.0745	1.9441	2.3251	2.8468	2.3720	1.1278	1.3225	1.5889	1.3464

Where $e_{m,c}$, $m = \{WTI, Brent, OPEC\}$, $c = \{95\%, 97.5\%, 99\%\}$ refer to the MSE for market m and confidence level c . $\overline{e_m}$, $m = \{WTI, Brent, OPEC\}$ refer to the average MSE for market m . It is clear that from results in Table 2, MSEs vary for different combinations of model parameters. These model parameters are used in CNN model to define the relevance of data used in the CNN model training. For example, for model parameters (2,2), the closes 2 observations such as the closest lag observations are selected during the model training process. Thus as the number get larger, more distance observations are convolved and transformed. The resulting feature map would represent the original data at a coarser level. In this paper, among different combinations of model parameters, we choose the model parameter that minimizes the MSE. Volatility is the key variable in the training of deep learning risk forecasting model using historical data. Unfortunately, volatility is an unobservable latent variable and needs to be calculated using some volatility measure. In

this paper, we use historical volatility as the volatility measure. It is calculated using the historical time window. Using the validation data, we have calculated historical volatility using different time windows such as 20, 90, 126, etc. We found 126 to be the optimal time window across both markets. Thus the time window for calculating the historical volatility is set to 126 trading days, which corresponds to a half year interval. The detailed network structures for CNN model in each precious metal market are listed in detail in table 3.

Table 3: Network structure of CNN model

Layer	KS_{WTI}	KS_{Brent}	KS_{OPEC}
<i>Conv</i>	6,2x2	7,2x2	7,2x2
<i>Pool</i>	1x1	1x1	1x1
<i>FC</i>	1	1	1

Where $Conv_i, i = 1, 2, \dots, n$ is the i^{th} convolutional layer, $Pool_i$ is the i^{th} max-pooling layer, KS is the kernel shape. FC_i is the i^{th} fully connected network.

Then we further conduct the empirical evaluation of the performance of the proposed models. Using the test data set, we further calculate the VaR at the confidence levels of 95%, 97.5%, and 99%, i.e., 5%, 2.5%, and 1% quantiles using both the proposed VMD-CNN model and the benchmark ARMA-GARCH model. ARMA-GARCH model is chosen as the benchmark model as it is the most widely used econometric model to forecast the volatility and risk measures in the literature. It produces the most robust forecasts. Meanwhile, although Artificial Intelligence models such as the Artificial Neural Network (ANN) and Support Vector Machine (SVM) models demonstrate the superior forecasting accuracy in a wide range of disciplines, they received less attention in the risk forecasting field. Different from the price return used in the forecasting field, volatility as the key variable in the risk forecasting field is an unobservable latent variable, which causes trouble for constructing the supervised training dataset since the training target is sensitive to the time windows used to calculate it. Besides, both ANN and SVM model are sensitive to the choice of network parameters. Since the parameter selection process lacks the theoretical guidance in the literature, these types of models also encounter the overfitting issue. Thus, the model performance is often trapped in the local minima and is not robust due to less optimal parameters chosen. Results

are reported in table 4.

Table 4: Model performance using the out-of-sample data

Models	$N_{95\%}$	$N_{97.5\%}$	$N_{99\%}$	$p_{95\%}$	$p_{97.5\%}$	$p_{99\%}$
$ARMA - GARCH_{WTI}$	28	18	9	0.0430	0.0118	0.0216
$VMD - CNN_{WTI}$	24	11	6	0.2421	0.6126	0.2879
$ARMA - GARCH_{Brent}$	30	20	9	0.0171	0.0027	0.0231
$VMD - CNN_{Brent}$	13	8	6	0.1323	0.6070	0.2982
$ARMA - GARCH_{OPEC}$	26	19	10	0.1363	0.0070	0.0088
$VMD - CNN_{OPEC}$	16	10	6	0.9432	0.9097	0.4277

From results in table 4, the VMD-CNN model has produced much better risk forecasts than the benchmark ARMA-GARCH model, indicated by the following performance evaluation. Firstly, VaRs produced by VMD-CNN model pass the backtesting procedure as the null hypothesis for the unconditional coverage test is not rejected and the p value is larger than the cut-off value of 0.05. In comparison, VaRs produced by ARMA-GARCH model failed to pass the backtesting procedure at all three confidence level in all three market, except for 95% confidence level in OPEC market. Secondly, the risk forecasts of the VMD-CNN model produce much lower number of exceedances. The VaRs estimated with VMD-CNN model are uniformly more conservative than VaRs produced by the benchmark ARMA-GARCH model.

In our study, the optimal performance for VMD-CNN model is achieved with the model parameter (2,6) in WTI market and (2,7) in both Brent and OPEC markets. It is clear that the optimal model parameters change when the model is fitted to different data sets. New set of optimal parameters is estimated when other market is involved in the investigation.

Our results show that the proposed algorithm in our model has been able to achieve the significant performance improvement in risk estimation. It is worth noting that the model performance has been achieved when the simple intuitive greedy search method is used to find the optimal parameters for the CNN model. The risk estimation accuracy can be further improved with the further improvement in the identification of the optimal parameters. The search for the optimal parame-

ters is an optimization problem involving the balance between the over fitting and generalization of the model performance. Many parameters need to be taken into account during the optimization process, such as the number of filters, the layer size, and the learning rate, etc. Besides, In this paper, both VMD model and CNN model are used to capture the multiscale and hierarchical data characteristics. But the multiscale and hierarchical data modeling is conducted in different manners in both models. In VMD model, the multiscale data features are captured and represented in the decomposed modes across different scales, with clear multiscale structure. The multiscale structure can be given some specific economic interpretation. In CNN model, the multiscale data features are captured during the training process using the filter. They are modeled in a series of data structure known as the feature maps when the data are filtered in the model. The feature maps at different layers represent the extracted and modeled information at different scales, from the more concrete to the more abstract levels. The extraction of this information, embodied in the network parameters, is carried out through training the CNN network using stochastic gradient descent algorithm, in a data adaptive and machine learning manner. It is usually much harder to assign the direct economic interpretations to these feature maps. Therefore, although VMD and CNN model both address the modeling of the multiple scales and hierarchical data structure, they are used in this paper with a complimentary relationship. VMD model takes an optimization approach to determine the optimal combination of decomposed modes. CNN model takes a data-adaptive and machine learning approach to reduce the biases in the risk forecasts and improve the risk forecasting accuracy. In VMD-CNN model, they work together to produce the multiscale based VaR forecasts with better economic interpretations and forecasting accuracy.

The improved risk forecasting accuracy from the VMD-CNN model suggests the necessity of the introduction of more complex model for estimating risk movement. In the literature, VMD based model typically assumes the linear and orthogonal relationship among risk estimates for different components extracted in the multiscale domain. But the practical data may not satisfy this linear assumptions. More complex models such as the CNN model may contribute to the modeling of more accurate nonlinear relationship between the individual scale level risk forecasts and the overall risk forecast.

4. Conclusions

In this paper we have proposed a new multiscale Value at Risk model based on the Variational Mode Decomposition model and the Convolutional Neural Network model. We used the VMD model to extract the main influencing factors and forecast their future risk movement. They are further modeled using CNN model to reduce the estimation biases and forecast the downside risk level. We have conducted the in-depth comparison of the performance of the proposed model using the extensive crude oil price data and found the positive evidence of the improved model performance.

Our results provide further evidence to the existence of the multiscale data structure as the important data feature underlying the crude oil price fluctuation. We find that to exploit the multiscale data feature, both multiscale analysis and CNN models could make significant contributions. There are several implications for the results in this paper. Firstly, we find that the transformation of the extracted data from the multiscale models to the two dimensional image data can reveal subtle details that are useful for modeling the hidden data features. Another major finding of this paper is that the CNN model can effectively capture the multi scale data characteristics and fully utilize them to improve the forecasting accuracy. Secondly, we find that the risk has multiscale hierarchical structure. This, combined with the success of the VMD-CNN model in the empirical forecasting exercises, implies that more advanced artificial intelligence models considering the data's multiscale hierarchical structure is expected to produce better modeling accuracy. The recently widely popular CNN model offers some promises in this aspect. The proposed model can be generalized to use other multiscale models and Artificial Intelligence models. The use of VMD model and CNN model in this paper provides a good starting point. This merit further exploration and modeling of the complex risk structure. Since we still have very little information and understanding about the complex risk structure and its movement, future research can be directed in this direction.

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***Declaration of Interest Statement**

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: